

Stellapps: System Architecture Document

Anjan Mahapatra

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Abstract

This document presents the comprehensive system architecture for **Stellapps**, India's leading full-stack dairy technology platform focused on IoT-enabled dairy farmer financing, cold chain infrastructure, and data-driven advisory. The architecture is designed to support **8 million+ farmers**, **10,000+ BMCs (Bulk Milk Coolers)**, **20 million+ daily IoT data points**, Rs.15,000 crore in AUM, and Rs.2,500+ crore in annual operating income by FY30F.

The architecture follows a **microservices-based, cloud-native design** with **offline-first capabilities** for rural connectivity and **IoT-first data ingestion**. Key components include: (1) IoT device management layer (mooMark smart meters, cold chain sensors), (2) multi-channel farmer/BMC onboarding system, (3) Stellapps Credit Engine for underwriting (84% prediction accuracy), (4) NBFC lending and collections module, (5) cold chain monitoring platform with real-time alerting, (6) parametric animal health insurance system, (7) data monetization platform for enterprise customers, (8) farmer advisory service via WhatsApp/SMS, and (9) analytics and reporting layer.

The architecture emphasizes **scalability** (horizontal scaling to 5M+ daily API requests), **fault tolerance** (99.9% uptime target, 4-hour RTO), **security** (AES-256 encryption, RBAC, mTLS for IoT devices), and **compliance** with RBI NBFC regulations, DPDP Act (India), and FSSAI quality standards. This document serves as the definitive reference for engineering teams, investors, and technology partners.

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1 Architecture Overview

Stellapps' system architecture is built on a **microservices-based, cloud-native design** deployed on AWS (Mumbai region) with edge nodes for offline operation in low-connectivity rural areas. The architecture follows the following design principles:

1. **Microservices-first**: Loosely coupled services that can be developed, deployed, and scaled independently
2. **Cloud-native**: Leveraging AWS managed services (EKS, RDS, S3, IoT Core) for reduced operational overhead
3. **Offline-first**: Local data persistence and sync capabilities for 30-40% rural areas with intermittent connectivity
4. **IoT-first**: Real-time ingestion and processing of telemetry from 600,000+ devices
5. **Security-by-design**: Encryption at rest and in transit, RBAC, comprehensive audit logging
6. **API-first**: Well-documented RESTful APIs for all external integrations

1.1 User Personas Supported

The architecture supports the following key user personas:

Persona	Capabilities
Farmers (individual)	Receive loans (milk-linked), track repayment schedule, access advisory (heat stress, mastitis alerts), file insurance claims, view milk prices
BMC operators/managers	Onboard farmers, record daily milk volume/quality, manage repayments, access Unnati SaaS platform, monitor IoT devices
Mitra/Field officers	Onboard farmers, conduct due diligence, collect repayments, provide training, install IoT devices
Dairy processors	Access farmer milk production data via API, place orders, integrate with Stellapps marketplace
Corporate users	Access data APIs, purchase insights (milk production trends, quality metrics), monitor portfolios
Stellapps admins	Underwrite loans, approve farmers, monitor KPIs, manage NBFC compliance, configure system

1.2 High-Level Architecture Diagram

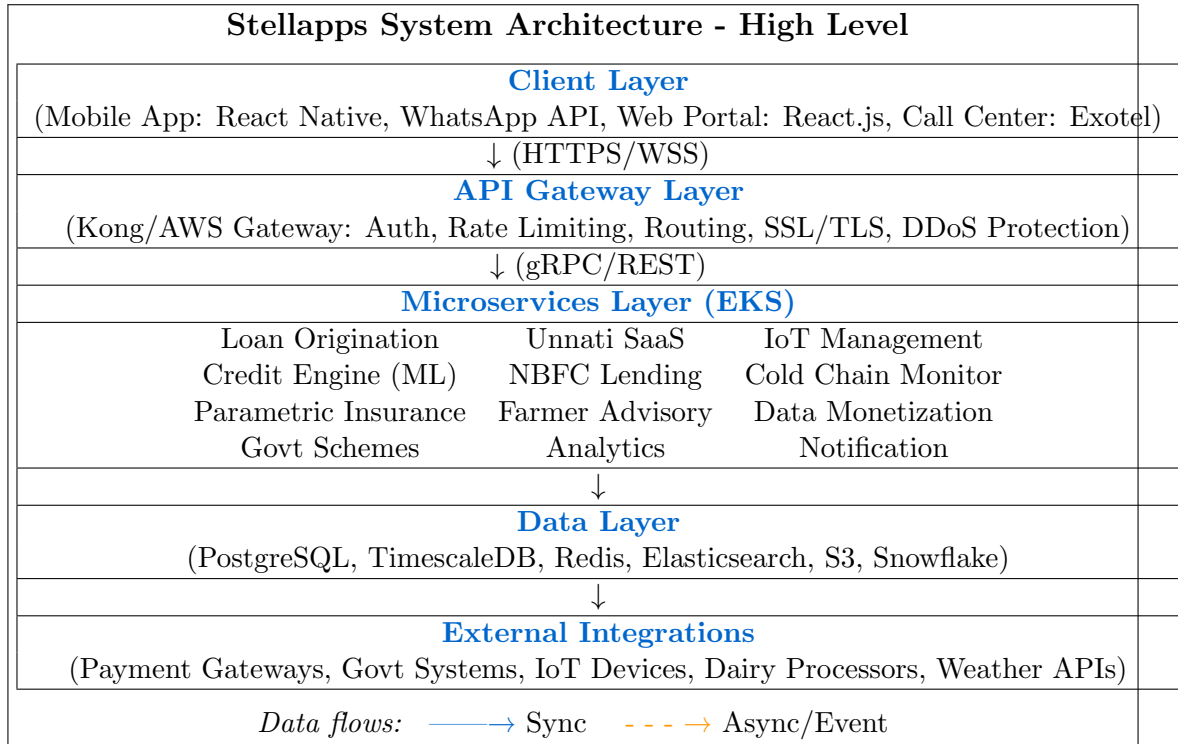


Figure 1: High-Level System Architecture

2 Client Layer

The client layer provides multi-channel access to Stellapps' services, catering to farmers and BMC operators with varying levels of digital literacy (30% smartphone penetration in rural India) and connectivity (40% areas with 4G coverage).

2.1 Mobile Application (Android & iOS)

- **Technology:** React Native (cross-platform), offline-first with SQLite local storage
- **App size:** 25 MB (optimized for low-end Android devices with 2GB RAM)
- **Minimum OS:** Android 8 (API 26) / iOS 13

2.1.1 Key Features by User Type

Table 2: Mobile App Features by Persona

Feature	Farmer	BMC Op	Field Officer
Loan application status			
Repayment schedule			
Advisory (heat stress, mastitis)		—	—
Milk prices		—	—
Insurance claim filing		—	—
Farmer management	—		
Milk volume/quality recording	—		—
IoT device status	—		
Onboarding checklist	—	—	
Offline mode			
Multilingual (10 languages)			

2.2 WhatsApp Integration

- **Technology:** WhatsApp Business API (Meta) via Cloud API
- **Monthly active users:** Target 2M+ by FY28
- **Use cases:**
 - **Loan status:** Farmer sends "STATUS [loan ID]" → automated response with outstanding amount, next due date
 - **Repayment reminder:** Automated notifications 7 days, 3 days, 1 day before due date
 - **Advisory:** Daily heat stress alerts (temperature $>35^{\circ}\text{C}$), mastitis detection tips, vaccination reminders
 - **Milk price:** Real-time procurement price from connected BMC
 - **Insurance claim:** File claim by sending "CLAIM [policy ID] [reason]"

- **Language support:** NLP for Hinglish and 10 regional languages using Meta’s multilingual model
- **Fallback:** Twilio SMS when WhatsApp unavailable or user opts out

2.3 Web Portal (Unnati SaaS for BMCs & Corporate Users)

- **Technology:** React.js with TypeScript, Tailwind CSS
- **BMC portal features:**
 - Farmer register management (Aadhaar-linked, biometric-enabled)
 - Daily milk procurement tracking (volume, fat/SNF, temperature)
 - Loan application and tracking
 - Repayment scheduling and payment gateway integration
 - IoT device dashboard (mooMark meters, cold chain sensors)
 - GST/TDS compliance and automated filing
 - Role-based access (admin, treasurer, operator)
- **Corporate portal features:**
 - API key management for data access
 - Data subscription management (tiered plans: Basic, Pro, Enterprise)
 - Portfolio analytics (for impact investors, government)
 - Custom report generation (CSV, PDF, Excel exports)

2.4 Call Center Integration

- **Technology:** Ozonetel / Exotel (cloud telephony)
- **Agent count:** 50+ agents by FY27, 150+ by FY30
- **Features:**
 - IVR (Interactive Voice Response) in 10 regional languages
 - Automated call-back for missed calls (farmer gives missed call → system calls back within 60 seconds)
 - Agent dashboard with customer 360° view (loan history, milk production trends, repayment status)
 - Screen pop with AI-suggested responses based on farmer’s past interactions (powered by LLM)
 - Call recording and quality assurance (random sampling for training)
 - Integration with CRM (Salesforce) for case management

3 API Gateway Layer

The API Gateway serves as the single entry point for all client requests, providing security, routing, rate limiting, and observability.

Table 3: API Gateway Configuration

Component	Specification
Technology	Kong (open-source) / AWS API Gateway
Authentication	JWT + OTP (farmers); API keys (corporate); OAuth2 (processors)
Rate Limiting	50/min (farmer); 500/min (BMC); 5,000/min (corporate); 10,000/min (internal)
Request Routing	Endpoint prefix: /loans, /farmers, /milk, /data, /insurance, /iot, /admin, /health
SSL/TLS	TLS 1.3 + auto cert renewal (AWS ACM)
DDoS Protection	AWS Shield Advanced + WAF (OWASP Top 10 rules)
Request Validation	JSON schema validation; size limits (1MB for farmer, 10MB for corporate)
Logging	Centralized to ELK stack; access logs retained 90 days
Caching	Redis cache for frequently accessed data (rate limit counters, JWTs)

3.1 API Versioning Strategy

- **URI versioning:** /v1/loans, /v2/loans
- **Deprecation policy:** Announce 6 months in advance; maintain backward compatibility for 12 months
- **Current versions:** v1 (stable), v2 (beta)

4 Microservices Layer

The microservices layer comprises loosely coupled services that can be scaled independently based on load. Services communicate via gRPC (synchronous) and Kafka (asynchronous events).

4.1 Microservices Summary

Table 4: Microservices Overview

Service	Language	Scaling Priority	Description
Loan Origination	Java Spring Boot	High	Loan lifecycle management
Credit Engine (SCE)	Python/ML	High	ML-based credit scoring
IoT Management	Python + MQTT	High	Device provisioning, telemetry
Cold Chain Monitor	Node.js	Medium	Temperature monitoring, alerts
Unnati SaaS	Python Django	Medium	BMC management platform
NBFC Lending	Java Spring Boot	High	Collections, RBI reporting
Parametric Insurance	Node.js	Low	Insurance policy management
Farmer Advisory	Python FastAPI	Low	WhatsApp/SMS advisory
Data Monetization	Python FastAPI	Low	Enterprise data APIs
Govt Schemes	Node.js	Low	DBT, subsidy processing
Analytics	Python + dbt	Medium	BI dashboards
Notification	Node.js	Medium	SMS, WhatsApp, email

4.2 Service 1: Loan Origination & Underwriting Service

This service handles the complete loan lifecycle from application to disbursement.

- **Technology:** Java Spring Boot 3.x, Drools (rules engine for loan structuring)
- **Database:** PostgreSQL (read-write) + Redis (caching for frequent queries)
- **API endpoints:** POST /loans/apply, GET /loans/id/status, POST /loans/id/disburse
- **Key functions:**
 - **Farmer onboarding:** Digital application with e-sign; Aadhaar-based KYC (UIDAI API); bank account verification (penny drop)
 - **Credit underwriting:** ML model using 50+ features (milk volume history, fat/SNF consistency, payment history, animal health)
 - **Loan structuring:** Tenor (6-24 months), interest rate (14-18%), repayment schedule (daily milk deduction)
 - **Disbursement:** Direct to farmer's bank account (via Razorpay Payouts) or as feed credit to verified suppliers
 - **Repayment tracking:** Automated deduction from milk payment (integration with BMC milk procurement system)

Table 5: Credit Scoring Model - Technical Specifications

Parameter	Value
Model architecture	XGBoost + LSTM (time-series)
Training data	2M+ farmer-months
Number of features	52
AUC-ROC	0.89
Default prediction accuracy (3 months)	84%
Inference latency (p95)	~500 ms
Feature freshness	Real-time (daily milk data)
Retraining frequency	Weekly (automated)

4.2.1 Stellapps Credit Engine (SCE) - Model Architecture

Feature Categories:

- **Milk Production (15 features):** Daily volume (6-month rolling), coefficient of variation, trend (increasing/decreasing), seasonality index
- **Milk Quality (12 features):** Fat%, SNF%, fat/SNF ratio, temperature at collection, conductivity (mastitis indicator)
- **Payment History (8 features):** On-time payment rate, average delay (days), max delinquency, payment consistency score
- **Animal Health (10 features):** Veterinary visit frequency, vaccination compliance, disease history, heat stress incidents
- **External Data (7 features):** Weather (heat wave days, rainfall), seasonal factors, local milk prices

4.3 Service 2: IoT Device Management Service

This service manages the fleet of mooMark smart milk meters and cold chain sensors.

- **Technology:** Python + MQTT broker (Eclipse Mosquitto) + AWS IoT Core
- **Protocols:** MQTT (telemetry), HTTPS (configuration)
- **Device count:** 85,000+ currently → 600,000+ by FY30F
- **Key functions:**
 - **Device provisioning:** OTA (Over-the-Air) configuration, firmware updates (signed binaries)
 - **Data ingestion:** Real-time ingestion of milk volume, fat/SNF, temperature (batched every 5 seconds)
 - **Device health monitoring:** Battery level, connectivity status, last heart-beat, tamper detection
 - **Anomaly detection:** ML-based detection of meter tampering (milk dilution, magnet interference, swapped milk)
 - **Command dispatch:** Remote configuration, calibration, reboot, firmware update

Table 6: IoT Device Telemetry Schema

Field	Type	Description
device_id	string	Unique device identifier (IMEI)
timestamp	datetime	UTC time of measurement
milk_volume	float	Litres (0.1L resolution)
fat_percent	float	Fat content (0.1% resolution)
snf_percent	float	Solids-Not-Fat (0.1% resolution)
temperature	float	Milk temperature at collection (°C)
conductivity	float	Electrical conductivity (mastitis proxy)
battery_voltage	float	Device battery level (V)
rssi	int	Cellular signal strength (dBm)
tamper_status	boolean	Tamper switch triggered

4.4 Service 3: Cold Chain Monitoring Service

This service monitors temperature and humidity across the cold chain (BMCs and transit vehicles).

- **Technology:** Node.js + MQTT + InfluxDB (time-series database)
- **Sensor count:** 500+ BMC units, 200+ transit vehicles → 2,000+ by FY30F
- **Key functions:**
 - **Real-time temperature monitoring:** BMC storage tanks (4°C target, 6°C alarm threshold), transit vehicles (temperature every 5 minutes)
 - **Alert generation:** SMS/WhatsApp alert when temperature exceeds threshold (≥6°C for ≥30 minutes) — average response time ≤10 seconds
 - **Quality prediction:** ML model predicts milk spoilage risk based on temperature time-series (AUC 0.85)
 - **Compliance reporting:** Automated FSSAI compliance reports for dairy processors (daily, weekly, monthly)
 - **Energy efficiency:** BMC compressor duty cycle analysis to identify inefficient units

4.5 Service 4: Unnati SaaS (BMC Management) Service

This service powers the Unnati platform used by BMCs for farmer management, milk procurement, and compliance.

- **Technology:** Python Django + Celery (async tasks) + PostgreSQL
- **Active BMCs:** 250 currently → 2,000 by FY30F
- **Key functions:**
 - **Farmer register:** Aadhaar-linked, biometric-enabled (fingerprint scanner integration), with milk production history (365-day rolling)

- **Milk procurement:** Daily volume/quality recording with barcode/RFID farmer identification, automated payment calculation
- **Loan management:** Application status, repayment schedule, interest calculation, delinquency tracking
- **Accounting:** Automated GST/TDS filing (GSTR-1, GSTR-3B), integration with Tally/Zoho via API
- **Compliance:** FSSAI reporting (monthly quality summaries), government scheme reporting (PM-KISAN, livestock subsidies)
- **Inventory management:** Feed inventory, veterinary supplies, equipment tracking

4.6 Service 5: NBFC Lending & Collections Service

This service manages the NBFC loan book, collections, and regulatory reporting.

- **Technology:** Java Spring Boot + Quartz (scheduler) + PostgreSQL
- **Loan book (AUM):** Rs.450 Cr currently → Rs.3,600 Cr by FY30F
- **Key functions:**
 - **Loan servicing:** Automated daily deduction from milk payment, late fee calculation (2% per month on overdue), delinquency tracking (30/60/90 days)
 - **Collections:** Automated SMS/WhatsApp reminders (7/3/1 days before due date); field officer assignment for overdue >30 days
 - **Early warning system:** ML model flags at-risk farmers (milk volume drop >20% + payment delay + health alerts) — accuracy 78%, 30 days ahead
 - **Regulatory reporting:** Automated RBI reporting (NPA classification, capital adequacy, concentration limits, ALM)
 - **Securitization:** Pool loan data for securitization transactions; investor reporting (monthly pool performance, delinquency rates)

4.7 Service 6: Parametric Animal Health Insurance Service

This service enables farmers to purchase animal health insurance with automated payout.

- **Technology:** Node.js + integration APIs (ICICI Lombard, HDFC Ergo, Bajaj Allianz)
- **Insurance penetration:** 2% currently → 30% by FY30F
- **Key functions:**
 - **Quote generation:** Real-time premium calculation based on animal type (cow/buffalo), age, breed, location (district-level risk factor)
 - **Policy issuance:** Digital policy document with QR code for verification; PDF stored in S3 with 7-year retention

- **Automated payout:** Trigger based on disease outbreak (district-level surveillance from Dept of Animal Husbandry) or temperature threshold ($>40^{\circ}\text{C}$ for 3+ days); payout credited to loan account within 7 days
- **Commission tracking:** Auto-accrual of 8-12% commission (paid monthly via Razorpay)
- **Claim settlement rate tracking:** Dashboard for insurer performance (target: $>90\%$ within 7 days)

4.8 Service 7: Farmer Advisory Service

This service delivers personalized advisory to farmers via WhatsApp and SMS.

- **Technology:** Python FastAPI + AI/ML (GPT-based LLM for localized content generation)
- **Advisory volume:** 10M+ messages/month by FY30F
- **Key functions:**
 - **Heat stress alerts:** SMS/WhatsApp alert when IMD forecast $>35^{\circ}\text{C}$ for 2+ consecutive days; suggests cooling measures (shade, water, ventilation)
 - **Mastitis detection:** ML model flags abnormal fat/SNF (drop $>0.5\%$) or conductivity ($>5\text{ mS/cm}$); sends alert "Your milk quality indicates possible infection. Consult vet."
 - **Breeding window optimization:** Based on historical estrus patterns and weather; predicts optimal breeding window (next 48 hours)
 - **Vaccination reminders:** Schedule and track vaccinations for each animal (FMD, lumpy skin disease, hemorrhagic septicemia)
 - **Feed optimization:** Recommended feed mix based on milk production (volume, fat%) and current feed prices

4.9 Service 8: Data Monetization Service

This service sells anonymized, aggregated data to enterprise customers via API.

- **Technology:** Python FastAPI + Apache Spark (data aggregation) + Snowflake
- **Data volume:** 10 TB raw data per month $\rightarrow$ 100 TB by FY30F
- **Key functions:**
 - **Data anonymization:** Remove PII (name, Aadhaar, mobile, GPS); aggregate at district/block level (minimum 100 farmers per cohort)
 - **API access:** RESTful APIs with API keys; rate limiting by subscription tier (Basic: 100 req/day, Pro: 1,000 req/day, Enterprise: 10,000 req/day)
 - **Report generation:** Scheduled reports (daily, weekly, monthly) for government (Ministry of Animal Husbandry) and feed companies (Cargill, Godrej)

- **Data quality monitoring:** Ensure 99.9% data accuracy; automated anomaly detection (sudden drops in volume, impossible fat%) with alerting
- **Datasets available:**
 - Milk production trends by region, breed, season (aggregated)
 - Quality metrics (fat/SNF) by region, month
 - Disease outbreak patterns and forecasts (anonymized)
 - Feed efficiency correlations (milk output vs. feed type)
 - Price realization by district, season

4.10 Service 9: Government Scheme Integration Service

This service integrates with government schemes to process payments and claim fees.

- **Technology:** Node.js + API integrations (NPCI, DBT, state govt portals)
- **Schemes supported:** PM-KISAN, livestock subsidies, feed subsidies, state-level schemes (10+)
- **Key functions:**
 - **Farmer enrollment:** Link farmers to livestock subsidies via Aadhaar; verify eligibility against government databases
 - **Subsidy processing:** Automatically credit subsidies to farmer loan account (reducing principal) or as cash to bank account
 - **Scheme reporting:** Generate reports for government on fund utilization, farmer reach, impact metrics
 - **Fee income:** Earn processing fee (1-2% of subsidy value) from government

4.11 Service 10: Analytics & Reporting Service

This service provides business intelligence and operational dashboards.

- **Technology:** Apache Superset / Power BI + dbt (data transformation) + Snowflake
- **Dashboard refresh:** Real-time (operational) / Daily (analytical)
- **Key dashboards:**
 - **Executive dashboard:** AUM, GNPA, ROA, ROE, disbursements by region, farmer growth, revenue trends
 - **Credit dashboard:** Loan application volume, approval rate (by region/score), average loan size, NPA trends (30/60/90+ days)
 - **Collections dashboard:** Repayment rate (on-time vs. delinquent), delinquency aging, field officer performance (by collector)
 - **IoT dashboard:** Device health (% online/offline), battery status distribution, tamper alerts by region

- **Cold chain dashboard:** Temperature compliance by BMC, spoilage alerts by route, energy efficiency ranking
- **Impact dashboard:** Farmers served (cumulative, monthly), women farmers percentage, income uplift (estimated)

4.12 Service 11: Notification Service (Cross-cutting)

This service handles all outbound communications across channels.

- **Technology:** Node.js + Bull (queue) + channel adapters
- **Channels:** WhatsApp (primary), SMS (fallback), Email (corporate), In-app (push)
- **Queue management:** Priority queues (high: OTP, repayment reminders; medium: advisory; low: marketing)
- **Delivery tracking:** Delivery status, open rates, click-through rates
- **Templates:** 50+ localized templates in 10 languages

5 Data Layer

The data layer consists of multiple storage systems optimized for different use cases (polyglot persistence).

5.1 Operational Databases

Table 7: Operational Databases

Database	Technology	Purpose	Size (FY30F)
Primary OLTP	PostgreSQL 15	Farmers, loans, repayments	20 TB
Time-series	TimescaleDB	IoT telemetry (milk volume, temp)	50 TB
Geospatial	PostGIS	BMC locations, farmer clusters	500 GB
Caching	Redis (cluster)	Session mgmt, rate limiting	200 GB
Search	Elasticsearch	Farmer/BMC search, logs	5 TB

5.2 Data Warehouse

- **Technology:** Snowflake (primary) / Amazon Redshift (legacy)
- **Data sources:** All microservices stream data to Kafka (50+ topics) → transformed via dbt (100+ models) → loaded to warehouse
- **Data volume:** 10 TB raw data per month currently → 100 TB by FY30F
- **Refresh frequency:** Real-time (streaming) for operational dashboards; daily for analytical reports (3 AM IST)
- **Schema:** Star schema with fact tables (loan facts, milk production facts, repayment facts) and dimension tables (farmer dim, date dim, location dim)

5.3 Data Lake

- **Technology:** Amazon S3 (with Glacier Deep Archive for cold storage)
- **Content:**
 - Raw IoT telemetry (milk volume, temperature, fat/SNF) — retained 90 days in S3 Standard (hot), 7 years in S3 Glacier Deep Archive (cold)
 - Farmer KYC documents (Aadhaar, land records, bank statements) — encrypted at rest (AES-256), retained 7 years post-last-transaction
 - BMC transaction logs (milk purchases, payments) — retained 7 years (RBI requirement)
 - System logs (application logs, access logs, audit logs) — retained 30 days in hot, 90 days in warm
 - ML model artifacts (XGBoost models, feature stores) — versioned, retained indefinitely

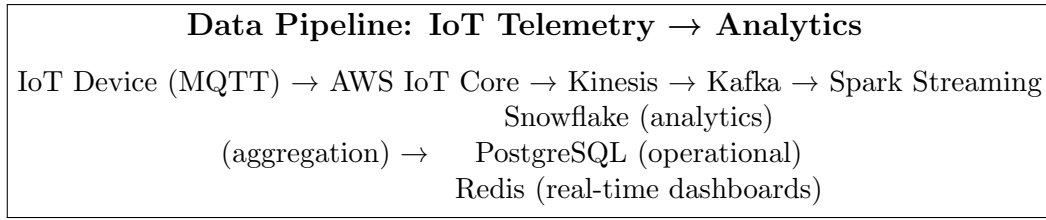


Figure 2: IoT Telemetry Data Pipeline

5.4 Data Pipeline Architecture

5.5 Data Governance & Security

- **Encryption:** AES-256 at rest (S3, RDS, Snowflake); TLS 1.3 in transit (all services)
- **Access control:** Row-level security in PostgreSQL; column-level security in Snowflake; IAM roles for AWS services
- **Data classification:** Public (milk price trends), Internal (operational metrics), Confidential (farmer PII), Restricted (loan details, Aadhaar)
- **Compliance:** RBI NBFC regulations (audit trail, data retention), DPDP Act (India), FSSAI (quality data), ISO 27001 (planned FY27)
- **Backup:** Automated daily snapshots (retained 30 days); cross-region replication to ap-southeast-1 (Singapore) for DR
- **Data lineage:** dbt docs for column-level lineage; Monte Carlo for data quality monitoring

6 External Integrations

Stellapps integrates with several external systems to enable seamless operations.

6.1 Payment Gateways

Table 8: Payment Gateway Integrations

Gateway	Use Case	Volume (FY30F)
Razorpay	UPI, cards, net banking for farmer repayments	Rs.500 Cr/mo
Cashfree	Farmer payouts (milk), field officer commissions	Rs.100 Cr/mo
Bank API (ICICI/HDFC/SBI)	Loan disbursement, NACH collection	Rs.200 Cr/mo

6.2 Government Systems

System	Integration Purpose
Aadhaar eKYC (UIDAI)	Farmer identity verification, age/address validation, biometric authentication
DigiLocker	Land record verification, animal registration certificate retrieval
GST portal (NIC)	Invoice validation for feed purchases and milk sales; GSTR-1/GSTR-3B filing
NDDB (Dairy cooperatives)	Data exchange for cooperative milk procurement, quality standards
IMD (Meteorological Department)	Weather forecasts (7-day), heat wave alerts, rainfall data
Dept of Animal Husbandry	Disease surveillance data (FMD, Lumpy Skin Disease outbreak alerts), vaccination schedules
DBT (NPCI)	Subsidy processing (PM-KISAN, livestock subsidies), beneficiary validation

6.3 Third-Party Services

- **Google Maps API:** BMC location geocoding, farmer distance calculation (for field officer route optimization)
- **AWS IoT Core:** Device connectivity (MQTT), message brokering, device shadow (state management)
- **Twilio:** SMS notifications (fallback when WhatsApp unavailable; 2M+ messages/month)
- **Sentry:** Error tracking and performance monitoring (real-time alerting on critical errors)
- **Datadog:** Infrastructure monitoring (CPU, memory, network, disk) and alerting (PagerDuty integration)

- **Dairy processor APIs:** Amul (NDDB API), Mother Dairy, Sid's Farm, Parag Milk Foods for milk offtake and price feeds
- **Cloudflare:** CDN for static assets (Unnati dashboard, mobile app downloads) and DDoS mitigation

7 Scalability & Performance

7.1 Scaling Strategy

The architecture is designed to scale horizontally to handle 10x growth by FY30F.

Table 10: Horizontal Scaling Targets by FY30F

Component	FY26F	FY30F	Growth
Active farmers (million)	2.8	8.0	2.9x
Active BMCs	500	2,000	4.0x
IoT devices deployed	150,000	600,000	4.0x
Daily IoT data points	2M	20M	10.0x
Daily loan applications	5,000	25,000	5.0x
Daily API requests	500,000	5,000,000	10.0x
Unnati SaaS active users	500	2,000	4.0x
Data warehouse size (TB)	10	100	10.0x

7.2 Auto-scaling Configuration

- **Orchestration:** EKS (Amazon Elastic Kubernetes Service) with Horizontal Pod Autoscaler (HPA) and Cluster Autoscaler
- **Trigger conditions:** CPU $\geq$ 70%, memory $\geq$ 80%, custom metric (requests per second $\geq$ 1,000, Kafka consumer lag $\geq$ 10,000)
- **Cooldown period:** 3 minutes (scale up), 10 minutes (scale down)
- **Node groups:**
 - **Compute-optimized (c5.large/c5.xlarge):** Loan origination, credit engine, NBFC lending — CPU-intensive workloads
 - **Memory-optimized (r5.large/r5.xlarge):** Redis caching, PostgreSQL, Elasticsearch — memory-intensive
 - **GPU-optimized (g4dn.xlarge):** ML model inference (credit scoring, anomaly detection) — 1-2 pods per node
 - **IoT-optimized (m5.large):** MQTT broker, telemetry ingestion, stream processing — network-optimized
 - **Spot instances:** Non-production environments, batch processing (data aggregation, report generation) — 60-70% cost savings

7.3 Performance SLAs

7.4 Performance Testing Strategy

- **Load testing:** JMeter (10,000 concurrent users, 5,000 req/sec target)
- **Stress testing:** Identify breaking points (2x peak load)
- **Endurance testing:** 24-hour sustained load at 80% peak

Table 11: Performance Service Level Agreements (SLAs)

Service	Target Response Time (p95)	Current (p95)
Loan application submission	<1 sec	0.8 sec
Credit decision (automated)	<5 sec	3.2 sec
IoT data ingestion	<500 ms	320 ms
Unnati dashboard load	<2 sec	1.5 sec
API data access (aggregated)	<1 sec	0.7 sec
Cold chain alert generation	<10 sec	5 sec
Mobile app sync	<3 sec	2 sec

- **Spike testing:** Sudden 5x traffic increase (flash sale, harvest season)
- **Frequency:** Quarterly (baseline), pre-release (every 2 weeks)

8 Deployment Architecture

8.1 AWS Infrastructure

Table 12: AWS Infrastructure Components

Component	Service	Configuration
Compute (K8s)	EKS	5-50 nodes (c5.large, r5.large, g4dn.xlarge)
Database	RDS PostgreSQL	Multi-AZ, db.r5.2xlarge (16 vCPU, 64 GB RAM)
Time-series	TimescaleDB	Managed on RDS, 5 TB storage
Caching	ElastiCache Redis	Cluster mode (3 shards, 2 replicas each)
Object storage	S3	100 TB standard, 500 TB Glacier
Data warehouse	Snowflake	X-Small to 3X-Large (auto-scaling)
Streaming	MSK (Kafka)	6 brokers (kafka.m5.2xlarge)
CDN	CloudFront	15+ edge locations in India
Networking	VPC	CIDR 10.0.0.0/16, 6 subnets (3 public, 3 private)

8.2 Primary Region: Mumbai (ap-south-1)

- **Production workloads:** All microservices, databases, data warehouse
- **Availability Zones:** 3 AZs (ap-south-1a, ap-south-1b, ap-south-1c) for high availability
- **Capacity:** 50 nodes (peak), 30 nodes (average)

8.3 Disaster Recovery

Table 13: Disaster Recovery Objectives

Metric	Target
Recovery Time Objective (RTO)	4 hours (critical systems: 1 hour)
Recovery Point Objective (RPO)	15 minutes (critical data: 5 minutes)
Backup frequency	Daily automated snapshots (RDS, EBS), continuous (S3 replication)
DR testing frequency	Quarterly (simulated region failover)
Warm standby	Core services (loan origination, IoT ingestion) running in DR region

DR Region: Singapore (ap-southeast-1)

- Cross-region replication for critical data (loan book, farmer profiles, IoT device registry)
- RDS read replica (async replication)
- S3 cross-region replication (CRR) for KYC documents and transaction logs
- Warm standby: EKS cluster with minimum nodes (20% capacity)

8.4 CI/CD Pipeline

- **Source control:** GitHub (monorepo with 15+ microservices)
- **CI:** GitHub Actions (build, test, lint, security scan)
- **CD:** ArgoCD (GitOps, automated deployment to EKS)
- **Environments:** dev → staging → production (canary deployment: 5% → 20% → 100%)
- **Rollback:** Automated (health check failure) or manual (one-click via ArgoCD)
- **Build frequency:** 50+ builds per day

9 Offline-First Design

Given that 30-40% of farmers and BMCs operate in areas with 2G/3G or intermittent 4G connectivity, Stellapps implements offline-first design principles for mobile applications.

9.1 Offline Capabilities by Component

Table 14: Offline Functionality Matrix

Component	Offline	Max Duration	Sync Strategy
Farmer onboarding	Yes	7 days	Batch sync (delta)
Loan application (draft)	Yes	30 days	Submit online (idempotent)
Milk volume recording	Yes	24 hrs	Priority queue upload
Repayment (field officer)	Yes	24 hrs	Upload with receipt
Advisory (cached)	Yes	7 days	Pre-download daily (5 AM)
Unnati dashboard	Partial	24 hrs	24h cache, stale-while-revalidate
Insurance claim (photos)	Yes	7 days	Upload on connectivity
IoT telemetry	Yes	30 days	Batch upload (MQTT)

9.2 Local Storage Architecture

- **Mobile app:** SQLite (up to 500 MB) for farmer data, loan applications, offline forms
- **IoT devices:** Flash storage (64 MB) for telemetry; store-and-forward for up to 30 days (10,000 readings)
- **Field officer tablets:** SQLite + file storage for KYC documents (photos, Aadhaar scans)

9.3 Sync Conflict Resolution

Table 15: Sync Conflict Resolution Strategies

Scenario	Resolution Strategy
Profile preferences	Last-write-wins (timestamp comparison)
Loan application	Server wins; client receives rejection with reason (idempotency key prevents duplicates)
Milk volume recording	BMC operator wins (trusted source); farmer input as dispute flag
Repayment collection	Server wins (financial transaction); field officer must reconcile offline
Disputed milk volume	Manual resolution by BMC manager; audit log maintained
Duplicate loan application	Idempotency key (UUID); second request returns existing application ID
KYC document upload	Server wins; client retries with exponential backoff (max 5 retries)

10 Security Architecture

10.1 Authentication & Authorization

Table 16: Authentication Methods by User Type

User Type	Method	Additional Controls
Farmers	OTP (mobile + Aadhaar)	OTP expires in 5 min; max 3 attempts
BMC operators	JWT + password	2FA optional; session timeout 8 hours
Field officers	JWT + geo-attendance	Must be within 500m of BMC for check-in
Corporate users	API keys + IP whitelisting	Key rotation every 90 days
Admins	MFA (TOTP) mandatory	Session timeout 2 hours; audit logging
Internal services	mTLS (service mesh)	SPIFFE identity; automatic cert rotation

10.2 RBAC Roles and Permissions

Table 17: RBAC Roles (8 roles)

Role	Permissions
Farmer	View own loans, repayments, advisory, insurance
BMC operator	Manage farmers at assigned BMC, record milk, view loans
BMC manager	All BMC operator permissions + approve loans, manage BMC settings
Field officer	Farmer onboarding, repayment collection, IoT installation
Credit officer	Loan underwriting, credit decision, limit setting
Admin	User management, system configuration, report access
Super admin	Full system access, audit log review, role assignment
Auditor	Read-only access to all data (compliance)

10.3 Security Controls Summary

Table 18: Security Controls

Control	Implementation
Authentication	OTP (farmers); JWT + MFA (admins); mTLS (services)
Authorization	RBAC (8 roles); ABAC for fine-grained (region, BMC)
API Security	JWT validation at gateway; rate limiting by role; input validation
Data Encryption (Rest)	AES-256 (RDS, S3, Snowflake, Elasticsearch)
Data Encryption (Transit)	TLS 1.3 (all services); mTLS for service-to-service
Secrets Management	AWS Secrets Manager (API keys, DB credentials, certs)
Vulnerability Scanning	Weekly automated (AWS Inspector, Snyk, Trivy)
Penetration Testing	Annual third-party (CERT-In empaneled firm)
Audit Logging	All API calls logged (user ID, timestamp, IP, endpoint)
IAM	Least privilege principle; temporary credentials (STS)
WAF	AWS WAF (OWASP Top 10, IP reputation, SQL injection)

10.4 IoT Device Security

- **Device identity:** X.509 certificates (per device) issued by AWS IoT Core
- **Mutual TLS (mTLS):** Device authenticates to cloud; cloud authenticates to device
- **Firmware signing:** OTA updates only with signed binaries; rollback protection
- **Secure boot:** Prevents unauthorized firmware
- **Tamper detection:** Physical tamper switch (triggered when enclosure opened)
- **Data validation:** Anomaly detection for impossible values (fat >15%, volume >100L)

10.5 RBI, FSSAI & DPDP Act Compliance

Table 19: Compliance Mapping

Requirement	RBI	DPDP	FSSAI
Data localization (India only)			—
Consent management			—
Data retention (7 years)		—	
Right to be forgotten	—		—
Data breach notification (72 hours)			—
Audit trail (all transactions)		—	
Quality reporting	—	—	
Encryption (AES-256, TLS 1.3)			

11 Monitoring & Observability

11.1 Monitoring Stack

Table 20: Monitoring Tools

Tool	Purpose
Datadog	Infrastructure metrics (CPU, memory, network)
Prometheus + Grafana	App metrics (latency, errors, business KPIs)
ELK Stack	Centralized logging (app, audit, access logs)
AWS X-Ray	Distributed tracing for microservices
Sentry	Error tracking
crash reporting	
PagerDuty	On-call alerting
incident management	
AWS CloudTrail	AWS API audit logging
AWS Config	Resource compliance monitoring

11.2 Key Metrics to Monitor

11.2.1 Business KPIs (Dashboard - 5 min refresh)

- Loan disbursement volume (daily, weekly, monthly by region)
- GNPA and NNPA trends (30/60/90+ days overdue)
- Repayment rate (on-time vs. delinquent, by region)
- Farmer onboarding rate (daily, weekly) and retention (78% target)
- Milk volume monitored (litres/day, by BMC)
- IoT device health (% online, battery >20%)
- Insurance penetration (% farmers insured)

11.2.2 Technical KPIs (Dashboard - real-time)

- API p95 latency by endpoint (target: <1 sec)
- Error rate (HTTP 5xx) — target <0.1% (99.9% availability)
- Credit engine inference latency — target <500 ms (p95)
- IoT data ingestion lag — target <1 second (from device to database)
- Database connection pool utilization (target <80%)
- Kafka consumer lag (data pipeline health) — target <10,000 messages
- Pod restart count (by service)
- Node CPU/memory utilization (target <70%)

11.3 Alerting Rules

Table 21: Critical Alerting Rules

Alert	Condition	Severity
API error rate >1%	5-minute rolling	P0 (page)
Credit engine latency >2 sec	5-minute rolling	P1 (page)
IoT ingestion lag >5 min	sustained 2 min	P1 (page)
Database CPU >80%	sustained 5 min	P1 (page)
Disk space <20%	—	P2 (email)
Pod restart count >5 in 10 min	—	P2 (email)

12 Cost Optimization

12.1 Current & Projected Cloud Spend

Table 22: Cloud Spend Projection (Rs. Cr)

Category	FY26F	FY28F	FY30F
Compute (EKS, EC2, Lambda)	4.0	8.0	15.0
Storage (RDS, S3, Snowflake, EBS)	2.0	4.0	8.0
IoT device connectivity (AWS IoT Core)	1.5	3.0	6.0
Data transfer (in/out, CloudFront)	0.5	1.0	2.0
Third-party APIs (maps, weather, SMS)	1.0	2.0	4.0
Total Cloud Spend	9.0	18.0	35.0

Note: Cloud spend as % of revenue declines from 7.5% (FY26F) to 1.4% (FY30F) due to operating leverage.

12.2 Optimization Strategies

1. **Reserved Instances (RI):** Purchase 3-year RIs for stable workloads (control plane, databases, Redis) → 35% savings on compute (Rs.1.4 Cr annually)
2. **Spot instances:** Use spot instances for non-production (staging, dev) and batch processing (data aggregation, ML training) → 65% savings (Rs.0.8 Cr annually)
3. **S3 lifecycle policies:** Transition older IoT data to S3 Glacier Instant Retrieval after 90 days, Glacier Deep Archive after 365 days → 80% storage cost reduction (Rs.1.6 Cr annually)
4. **CloudFront caching:** Cache Unnati dashboard assets (React bundles, images) and mobile app downloads at edge → 70% reduction in origin fetches (Rs.0.3 Cr annually)
5. **Data compression:** Compress IoT telemetry (gzip, 70% compression) and log data before transmission → 50% reduction in data transfer costs (Rs.0.2 Cr annually)
6. **Serverless for low-traffic services:** Use Lambda for infrequent operations (report generation, data exports, nightly batch) → pay-per-execution (Rs.0.1 Cr vs. Rs.0.5 Cr for EC2)
7. **IoT data aggregation:** Aggregate telemetry at edge (device) before cloud upload → 40% reduction in ingestion costs (Rs.0.6 Cr annually)
8. **Graviton (ARM) instances:** Migrate compute workloads to AWS Graviton (c7g/r7g) where compatible (Java, Python, Node.js) → 20% better price-performance

12.3 Cost Monitoring & Governance

- **Budget alerts:** Monthly budget thresholds (75%, 90%, 100%) via AWS Budgets
- **Tagging strategy:** Resource tags (service, environment, cost-center) for granular cost allocation

- **Monthly review:** Cloud cost review meeting (Finance + Engineering), optimization tickets tracked in Jira
- **Anomaly detection:** AWS Cost Anomaly Detection (ML-based), alerts for >15% daily spend increase

13 Conclusion

The Stellapps system architecture is designed to support the company's ambitious growth targets from Rs.80 crore (FY24A) to Rs.2,500+ crore (FY30F) while maintaining high availability (99.9% uptime target), sub-second response times for critical operations (loan application, credit decision), and full compliance with RBI NBFC regulations, DPDP Act (India), and FSSAI quality standards.

13.1 Key Architectural Decisions Summary

Table 23: Key Architectural Decisions & Rationale

Decision	Rationale
Microservices over monolithic	Independent scaling of IoT ingestion (high volume) vs. loan origination
EKS (Kubernetes) over ECS	Portability, larger community, Horizontal Pod Autoscaler
PostgreSQL + TimescaleDB	Single database for transactional + time-series (IoT telemetry)
MQTT for IoT telemetry	Lightweight, persistent sessions (offline support), QoS levels
Offline-first mobile apps	30-40% rural areas with intermittent connectivity
Snowflake for analytics	Separation of compute/storage, auto-scaling, zero-copy cloning
Kafka for event streaming	Decouple microservices, replay capability, 50+ topics
React Native over native	Single codebase for Android/iOS, faster feature rollout

13.2 Projected Capacity by FY30F

- **Farmers:** 8 million+ (5.3x growth from 1.5M)
- **BMCs:** 2,000+ (8x growth from 250)
- **IoT devices:** 600,000+ (7x growth from 85K)
- **Daily IoT data points:** 20 million (10x growth)
- **Daily API requests:** 5 million (10x growth)
- **AUM:** Rs.15,000 crore (44x growth from Rs.340 Cr)

- **Annual revenue:** Rs.2,500+ crore (31x growth)

The architecture is projected to handle these targets with:

- **Uptime:** 99.9% (8.76 hours downtime/year)
- **P95 latency:** <1 sec for critical APIs
- **RTO:** 4 hours (DR)
- **RPO:** 15 minutes (5 min for critical data)

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